

T. Ben Baker, PhD, JD

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ACADEMIC APPOINTMENTS

- **Colby College, Assistant Professor** 2023–present
Waterville, ME
– Department of Philosophy, Davis AI Institute
- **University of Pennsylvania, Provost Postdoctoral Fellow** 2020–2023
Philadelphia, PA
– Kording lab, Departments of Bioengineering and Neuroscience

EDUCATION

- **University of Pennsylvania, PhD Philosophy** 2020
Philadelphia, PA
– Dissertation: “Cognition in Nature: Information, Explanation, and Embodiment”
– Social, Cognitive and Affective Neuroscience, Graduate Certificate
– Dean’s award for distinguished teaching by a graduate student
- **Yale Law School, JD** 2014
New Haven, CT
– Substantial Analytic Writing on Mens Rea and Manslaughter
– Lowenstein International Human Rights Clinic
- **Brown University, BA (Concentrations in Philosophy (honors) and Economics)** 2010
Providence, RI

PUBLICATIONS

Peer-Reviewed Articles & Book Chapters

- **Use and Usability: Concepts of Representation in Philosophy, Neuroscience, Cognitive Science, and Computer Science** [\[link\]](#) 2026
with R Lange, A Richmond, R Cao, N Kriegeskorte, O Schwartz, X Pitkow, Neurons, Behavior, Data Analysis, and Theory
- **Rethinking Artificial General Intelligence: Beyond Anthropomorphism** forthcoming
Invited book chapter, Introducing Philosophy of Mind, Today (eds. D Curry, L Daoust), Routledge
- **Understanding Agent Complexity Through Affordances** [\[link\]](#) 2025
with Sonia Roberts, Minds and Machines
- **Computational Kinematics of Dance: Distinguishing Hip Hop Genres** [\[link\]](#) 2024
with T Liu, J Matelsky, F Parodi, B Mensh, J Krakauer, K Kording, Frontiers in Robotics and AI
- **Three Aspects of Representation in Neuroscience** [\[link\]](#) 2022
with B Lansdell and K Kording, Trends in Cognitive Sciences
- **Creativity in AI-Dance Art** [\[link\]](#) 2022
Proceedings of the International Conference on Computational Creativity
- **Natural Information, Factivity, and Nomicity** [\[link\]](#) 2021
Biology & Philosophy
- **Referee Report of (hypothetical) Philosophy101 Textbook** [\[link\]](#) 2021
Teaching Philosophy
– Spanish translation by Fredy H.P. Galindo, 2023, *Cuestiones de Filosofía*

Manuscripts in Progress

- **Six Dimensions of Dance Cognition**

in preparation

Philosophy of cognitive science article

- An account of six distinct agent-level abilities engaged in dance — articulated primarily through Black social dance styles — offering dance scholarship a tool for analyzing the thinking in different practices, and cognitive science a detailed map of an activity often invoked to show what mind owes to body

- **Felt, Seen, and Measured**

in preparation

Self-assessment, observer perception, and high-speed kinematics of learning a dance phrase

- Eleven dancers learned a hip-hop phrase from its choreographer and performed it under a four-camera high-speed markerless motion-capture rig; the paper maps where felt improvement, observer judgments, and measured kinematics (deviation, timing, amplitude, predictability) agree and diverge

Public Writing & Reports

- **Commentary on Favela and Machery** [\[link\]](#)

2023

Symposium contribution, The Brains Blog

- **The Time is Ripe to Reverse Engineer an Entire Nervous System** [\[link\]](#)

2023

Position paper, Alliance for C. elegans Neural Dynamics (ACeND)

- **What Does Engaged Philosophy Look Like?** [\[link\]](#)

2016

with L Daoust and R Willison, guest post, DailyNous.com

- **Before Their Time: Challenges to the Prohibition Against Child Marriage in Sierra Leone** [\[link\]](#)

2013

with A Elbogen, S Keene, Allard K. Lowenstein International Human Rights Clinic

TEACHING

Instructor of record for 17 semester-courses (300+ students through Spring 2026). Mean instructor rating 4.45/5 across evaluated sections. Frequency in parentheses where > 1.

Courses

- **Colby College, Assistant Professor, Department of Philosophy**

2023–present

Waterville, ME

- All courses developed and first-taught by me
- Central Philosophical Issues: Mind, Body & Machine (×6)
- Philosophy of Dance (×2)
- AI and Human Values
- Philosophy of Emotion
- Philosophy of Cognitive Science
- Philosophy of Science
- Bodies and Power

- **University of Pennsylvania, Instructor of Record, Department of Philosophy**

2018–2020

Philadelphia, PA

- Introduction to Philosophy (×2)
- Philosophy and Visual Perception
- Ethics

Student Advising

- **Honors theses:** 1 in Philosophy; 1 in Computer Science (cross-departmental)
- **Senior independent studies:** 2 in Philosophy
- **January-term and summer mentees:** 4 students

Other Teaching

- **Davis AI Summer Bridges** (Summer 2026, *Colby*): Guest instructor for a full-day session with 30 students of mixed coding experience — analysis of 3D dance-motion data with interpretable movement features, a hip-hop dance session, and a technology-ethics discussion

TALKS

Invited Talks

- **Commentator, Symposium Session** 2027
American Philosophical Association, Eastern Division, Boston MA
– Commentary on J Yi, “The Responsible Architect: An Existential-Hermeneutic Ethics for AI in Design”
- **Commentator, “AI, Stories and Theories”** 2026
American Philosophical Association, Eastern Division, Baltimore MD
– Commentary on I Oved et al., “Virtual Embodied AI for a More Rigorous Philosophy and Science of the Mind”
- **What Does It Mean to Be Human?** 2025
Invited all-school speech, Loomis Chaffee School, Windsor CT
- **Rethinking Artificial General Intelligence** 2025
Oberlin Colloquium in Philosophy, Oberlin College
- **Invited Commentator, Symposium on Technology and Cognition** 2025
Southern Society for Philosophy and Psychology, annual meeting
- **Moving Between Genres: A Computational and Embodied Exploration of Dance** 2025
Invited contributor, Slippage: 3D Humanities Series, Northwestern University
- **Dance Machines!** 2024
James Madison University, sponsored by the Philosophy & Religion Department and JMU Dance Program
- **Six Forms of Dance Cognition** 2023
Mind the Body, University of Utah

Conference Presentations & Panels

- **Ethics of AI-Mediated Language Education: Critical, Ecological, and Technomoral Perspectives** 2026
with C Wang, Artificial Intelligence Research in Applied Linguistics (AIRiAL) Conference, New York NY
- **So You Think AI Can Dance?** 2022
Eye, Mind, and Image: Themes from Visual Studies, UPenn
- **Panelist, Symposium on Affirming Black Excellence in Neuroscience** 2021
Cognitive Neuroscience Society (virtual)
- **Representation in Neuroscience** 2020
Neuromatch Conference 3.0 (virtual)
- **The Factivity and Nomicity of Information** 2019
Southern Society for Philosophy and Psychology, Cincinnati OH
- **Neural Dynamics and Inner Representations** 2017
Conference on Representing Reality, SUNY Potsdam
- **Learning to Dance with Dynamic Systems** 2017
Graduate BEN Talks, UPenn (internal)
- **Weakness of Will in Desire-First Accounts of Rational Action** 2015
Philosophers’ Cocoon Conference, Tampa FL

COMPUTATIONAL KINEMATICS OF HUMAN MOVEMENT

- **Motion Capture & Data Collection (Kinelab)** 2024–present
Colby College
– Designed and built a markerless motion-capture system at Colby; data collection ongoing in collaboration with Veronica Romero (Colby Psychology) under a joint umbrella IRB
- **Media Coverage**
Computational Kinematics of Dance (Frontiers in Robotics and AI, 2024)
– *Dance Magazine*, feature on AI, motion capture, and dance [\[link\]](#)
– *Colby Magazine*, feature on the dance-genre classifier
– SciTube, educational video episode on the paper

- **Movement Modeling (KineTrace)** 2025–present
Colby College
– Developing self-supervised next-window-prediction models trained on a custom biomechanical featurization of human movement (precursor: *Frontiers* 2024); early-stage collaboration with Amy LaViers (Robotics, Automation, and Dance Lab, Germantown PA)
- **Responsible AI Grant (\$10,000)** 2026
Davis Institute for Artificial Intelligence, Colby College
– Supporting sabbatical research on responsible computational kinematics — algorithmic bias, interpretability, and movement privacy in markerless motion capture
- **Technical Working Group, NEA Research Lab** 2026–present
Emory University (PI: Madeleine E. Hackney)
– Advisory group for a National Endowment for the Arts Research Lab on arts-based rehabilitation for people with Parkinson’s disease and dementia — intergenerational partnered dance, and gait, timing, and rhythmic-perception training; with Science Gallery Atlanta and Rockefeller University

PROFESSIONAL SERVICE & LEADERSHIP

- **Departmental/College Service, Colby College** 2023–present
 - **Department of Philosophy** (2023–present): Clark Award and Pepper Prize judging (reading and scoring senior essays, coordinating prize text, presenting at the awards banquet); co-organizing the annual Philosophy Department Awards Banquet; reviewing honors-thesis proposals for the Honors Program
 - **Faculty Advisory Committee, Davis Institute for Artificial Intelligence** (2023–present): Planning around Colby’s AI policies and tools, event organization
 - **Reuman Reading Group Organizer** (2024–present): communications, documentation, and publicity for Colby’s 27-year-running faculty-student reading group in philosophy
 - **Search Committees** (2023–2026): three for hires in Philosophy (one TT, two VAP); one for the Director of the Davis AI Institute; one as external reviewer for Psychology
 - **Lunder Institute for American Art, Faculty Advisory Group** (2024–2025): Planning and participating in initiatives around integrating museum resources and visiting fellows into the work and teaching of Colby faculty, handling internal grant funding
 - **Science, Technology, and Society Faculty Advisory Board** (2023–2025): Contributing to initiatives for the STS department around student participation, course options, and internal funding
- **Conference/Event Organization,** 2018–present
 - **Philosophy Symposium** (Spring 2025, *Colby*): Organized annual public lecture on Ethics of Social AI, with student participation, sponsored by the Philosophy department and the Davis AI Institute
 - **The Flavor Continues Residency Organizer** (2024, *Colby*): Coordinated residency for a Boston-based Hip Hop non-profit, offering community workshops, guest teaching in Colby dance classes, and a performance
 - **Birge Lecture Co-organizer** (Fall 2023, *Colby*): Helped select speaker and support publicity for Colby’s annual Kingsley Birge Memorial Lecture in Sociology
 - **Philosophy and Engagement Conference Co-organizer** (2019, *UPenn*): Collaborated on planning and executing an interdisciplinary conference exploring the intersection of philosophy and community engagement
 - **Indian Philosophy Reading Group** (2019, *UPenn*): Managed communications around logistics and selecting readings for graduate student discussion group
 - **Admitted Student Weekend Coordinator** (2018, *UPenn*): Organized accommodations, planned activities, and facilitated interactions between faculty and prospective graduate students during recruitment weekend
- **Peer Review,** ongoing
 - **Journal reviewer:** *Philosophical Psychology*; *Philosophy of Science*; *IEEE Robotics and Automation Magazine*; *Dance Research Journal*